## (a) Overview

### Support-set adaptation · once per support set

**Support set S**
**2K labeled images**
*K real + K fake*
Real
Fake

*Learn P*

CLIP ViT-L/14 blocks 1–24 ❄

image feature

Cosine logits
$S = exp(logit_{scale})$

logits

CE ← *y*

Fixed text anchors $t_0$, $t_1$ ❄

**Backprop through frozen encoder**

**Update prompts only**

CLIP ViT-L/14 blocks 1–12 ❄

normalized features

**Support cache**
… … $z_{ci}$, $\mu_c$

After optimization

**Fitted prompts P***

8,192 trainable parameters

### Query inference

**Query set Q**

*x ∈ Q*

CLIP ViT-L/14 blocks 1–12 ❄

$q \in \mathbb{R}^{1024}$

**Query-conditioned prototype mixing**
detail in (b)

$p_P(q)$

cache

**Stage-wise visual prompting**
detail in (c)

*Load P**

CLIP ViT-L/14 blocks 1–24 ❄

$v(x) \in \mathbb{R}^{768}$

**Cosine similarity difference**
*×15 → sigmoid*

Fixed text anchors $t_0$, $t_1$

$p_V(x)$

**Probability average**
$p(x) = ½[p_P(q) + p_V(x)]$

*p(x) > 0.5: Fake*

## (b) Query-conditioned prototype mixing

**Class-wise softmax** · **Weighted sum** · **Mix → L₂ norm**

c = 0
c = 1
*q*
Σ
$m_0(q)$
$m_1(q)$
$\mu_0$
$\mu_1$
0.5
0.5
+
$L_2$ norm
$\tilde{\mu}_0(q)$
$\tilde{\mu}_1(q)$

$\alpha_{ci}(q) = softmax_i(15\ q^T z_{ci})$

$m_c(q) = \Sigma_i \alpha_{ci}(q)\ z_{ci}$

$m_c(q)$ is not normalized before mixing.

$p_P(q) = \sigma(15\ q^T(\tilde{\mu}_1(q) - \tilde{\mu}_0(q)))$

## (c) Stage-wise prompt replacement

Frozen backbone ❄

$P_1$* $P_2$* $P_3$* $P_4$*

1–6 → 7–12 → 13–18 → 19–24

CLS readout project $L_2$ norm
*v(x)*

CLS + image tokens continue

**Keep CLS/image → discard prompt states → insert P***

CLS · prompt states · image → $P_{s+1}$* → Next stage

discard

*8,192 trainable parameters*

Real · Fake · Trainable / fitted prompts · ❄ Frozen parameters